

Practice Lab:

Install, configure and use
Eclipse

- This Practice Lab has no specific preceding Discussion Unit.
- In this Practice Lab we install, configure and use Eclipse, so that you can edit, compile and run Java client programs that target the DSE database server. As such, this lab expects an operating DSE cluster be accessible.
- Instructions in this practice lab are written for MacOS, but should function for most Linux distros.
- This Practice Lab will require reasonable Internet access.

Eclipse IDE: What is it ?

Open source integrated developer's workbench (IDE)

- From 1984's Visual Age, a similar IBM program
- Not apache, is Eclipse Public License (ECL)
- Releases with science related names
- June/2018 release is Photon
- Many/many development languages supported
- Extensible (Vi plugin !)



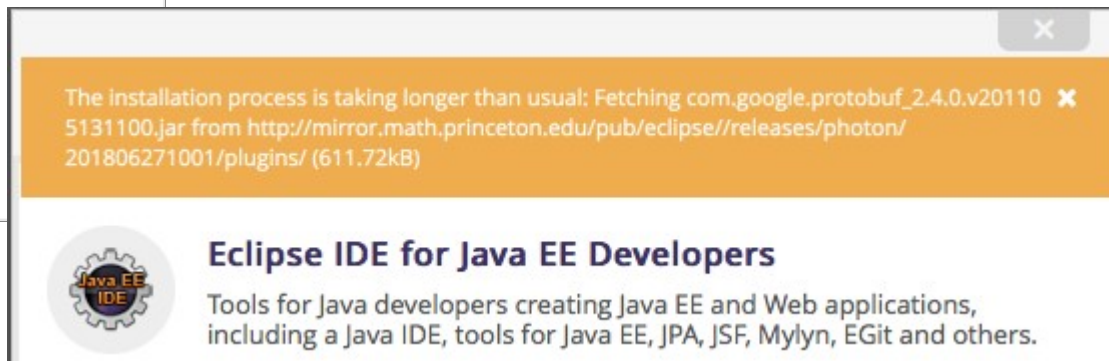
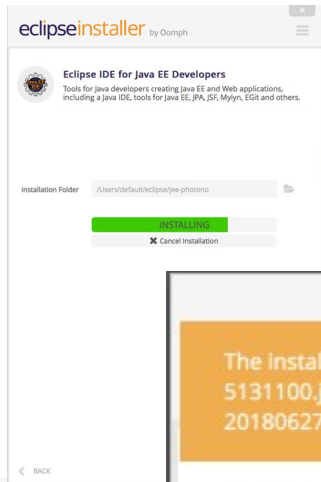
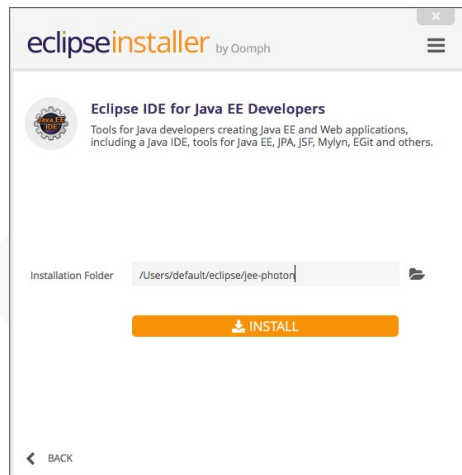
Eclipse: How to Install



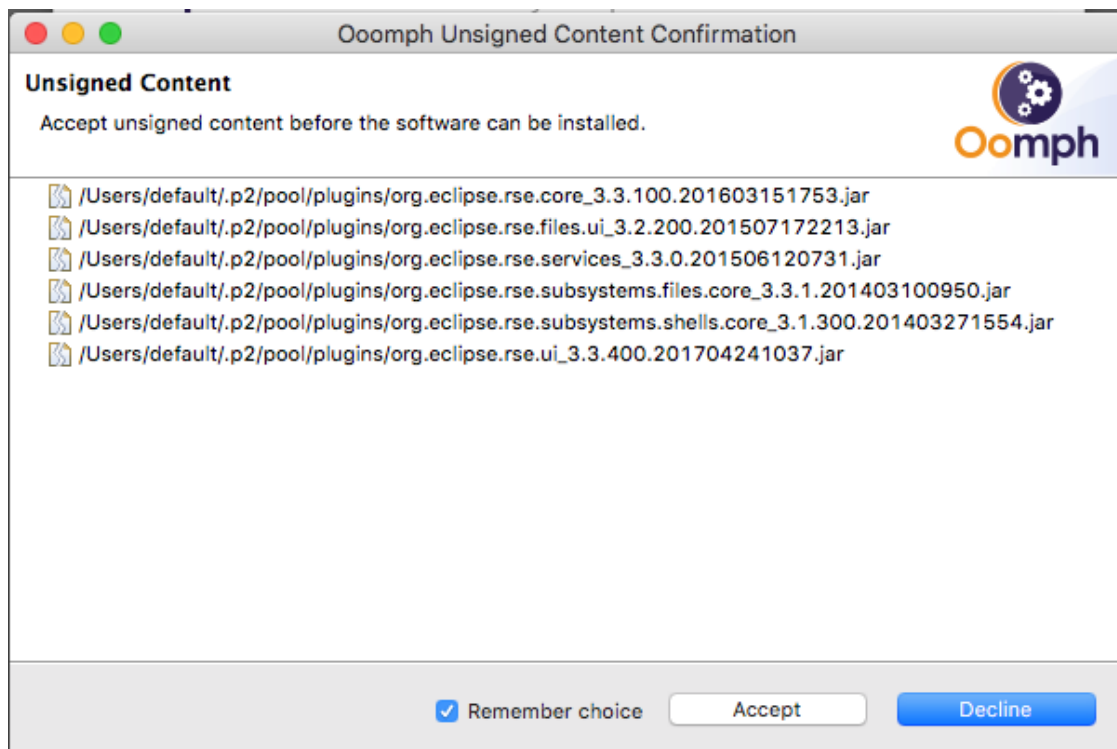
**Instructions that follow
are for MacOS, and
Eclipse Photon**

- Installer program, 49MB,
<http://www.eclipse.org/downloads/>
- Tar ball, yields DMG file
- Double-Click the DMG
- Choose, "Eclipse IDE for Java EE Developers"
- And Click, Install

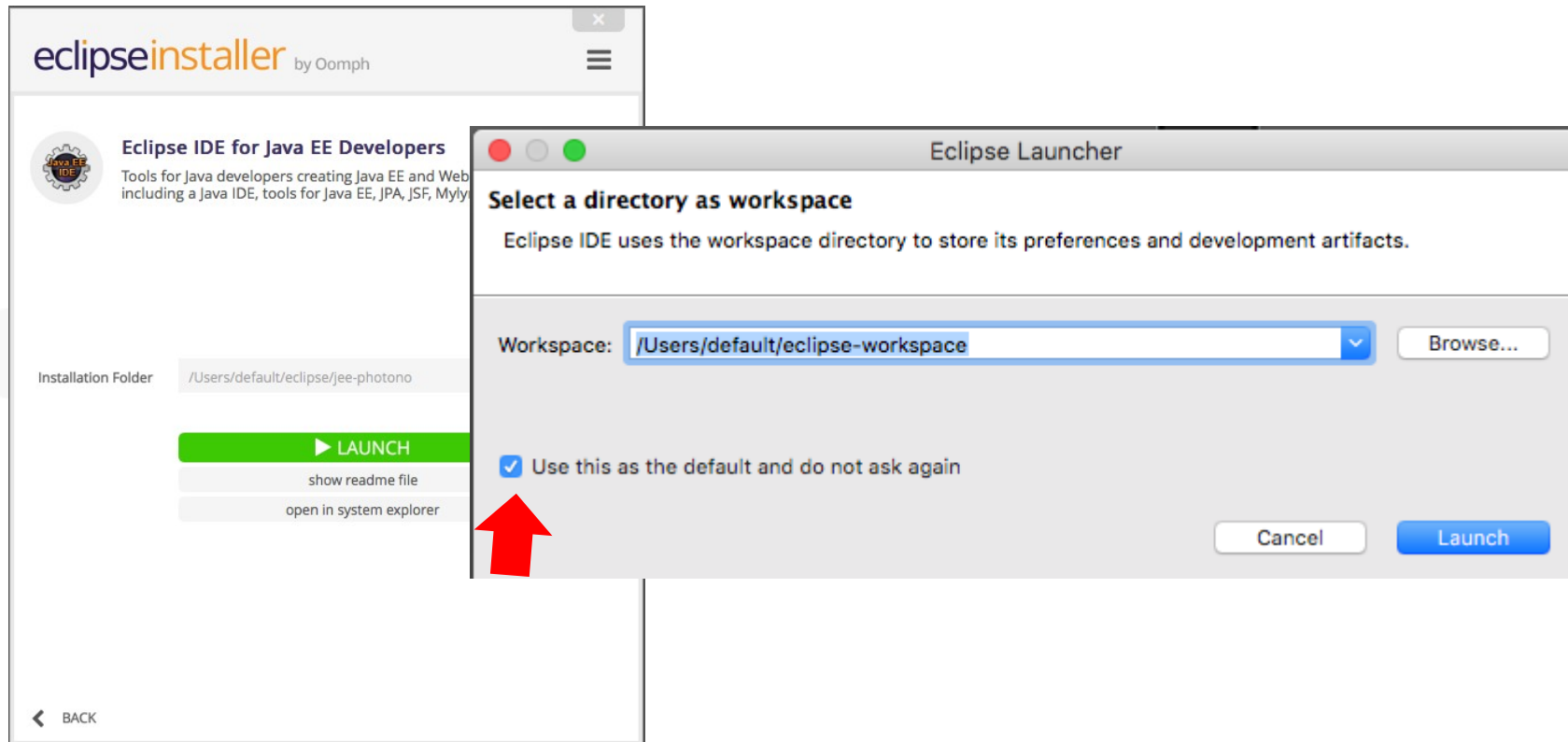
Eclipse Install: Spurious Errors



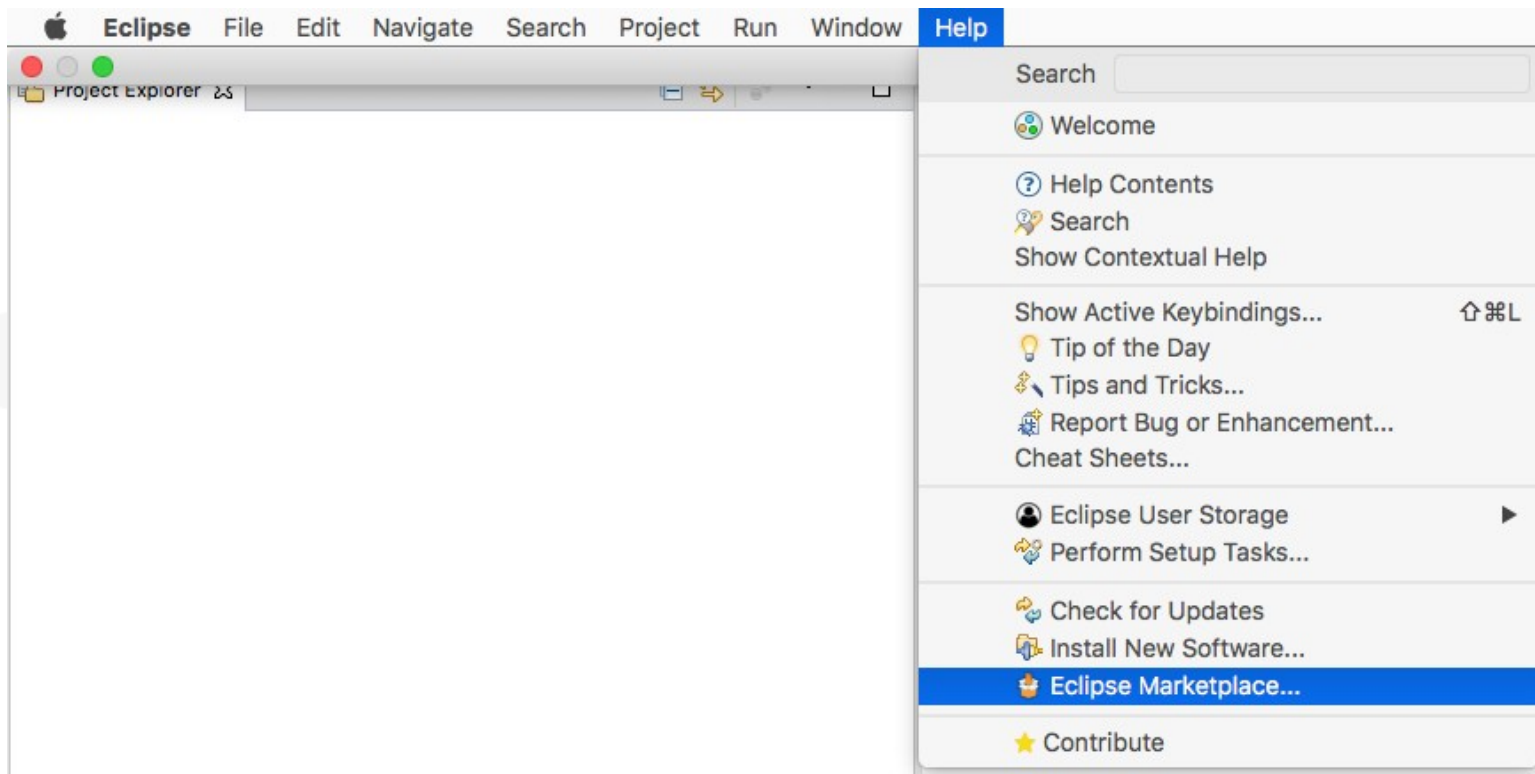
Eclipse Install: Accept Licenses



Eclipse: Launch, and Workspace



Eclipse: Install Scala plugin



Eclipse: Adding Scala IDE 4.7.x

The screenshot shows the Eclipse Marketplace interface. At the top, there's a search bar with 'scala' entered. Below the search bar, the 'Welcome to the Eclipse Marketplace' section is visible. To the right, a detailed view of the 'Scala IDE 4.7.x' plugin is shown. This view includes the Scala IDE logo, a description of the plugin, its version (4.7.x), and an 'Install' button. Red arrows highlight the search bar, the 'Welcome' section, and the 'Install' button.

Eclipse Marketplace

Select solutions to install. Press Install Now to proceed with installation.
Press the "more info" link to learn more about a solution.

Search Recent Popular Favorites Installed Eclipse Newsletter: Boot & Build Eclip...

Find: All Markets All Categories Go

Welcome to the Eclipse Marketplace

If you already have some favorite Marketplace entries, now is the perfect time to install them.

Featured

Vaadin Plugin for Eclipse 4.0.2

Promoted - Vaadin Framework is an open source Java UI library for creating rich web user interfaces. Using its component based API developers can create stunning web... [more info](#)

by [Vaadin Ltd](#), Apache 2.0
[java](#) [J2EE](#) [web](#) [ria](#) [java ee](#) ...

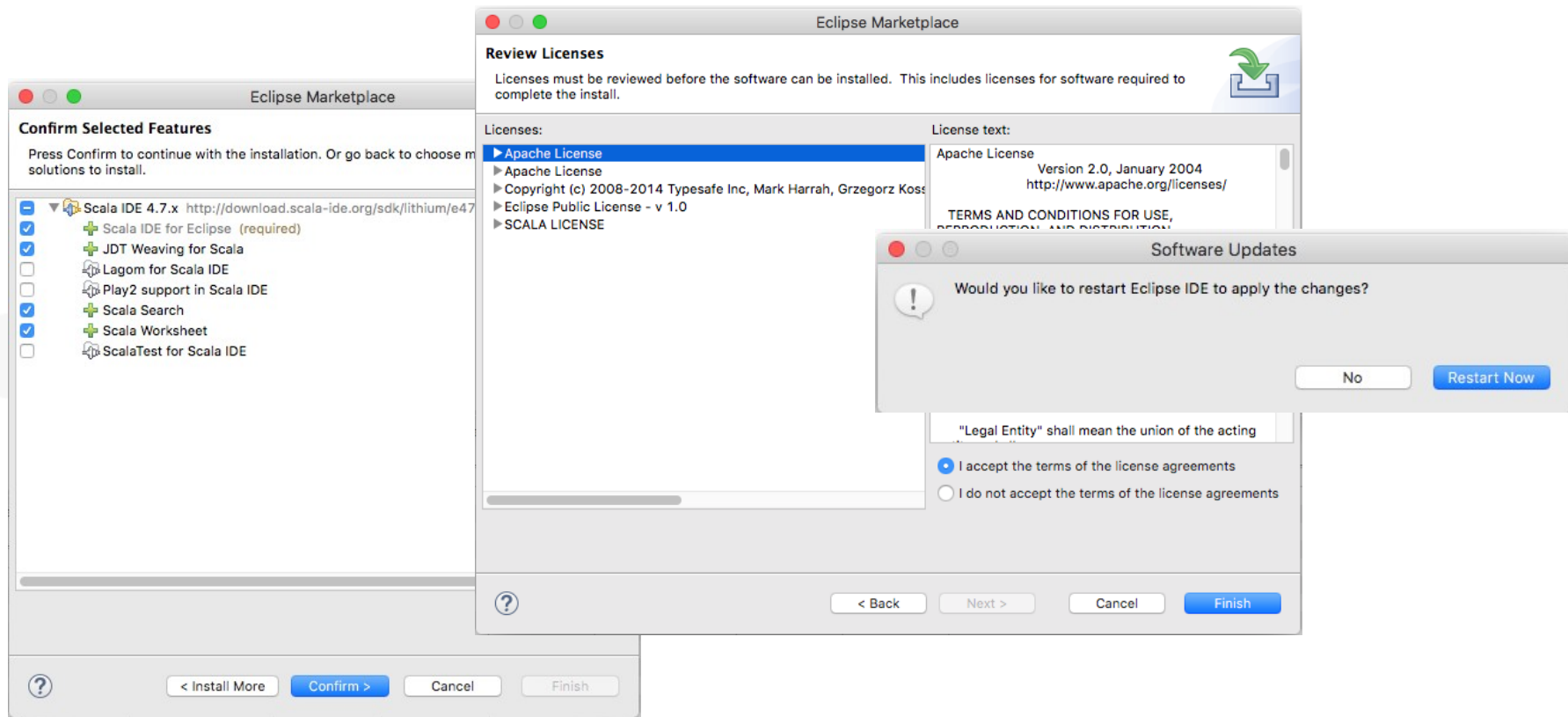
Scala IDE 4.7.x

The Scala IDE for Eclipse is centered around seamless integration with the Eclipse Java tools, providing many of the features Eclipse users have come to expect... [more info](#)

by [scala-ide.org](#), BSD
[scala](#) [functional programming](#) [fileExtension_scala](#)

★ 179 Installs: 200K (3,643 last month) **Install**

Eclipse Install: Accept Licenses

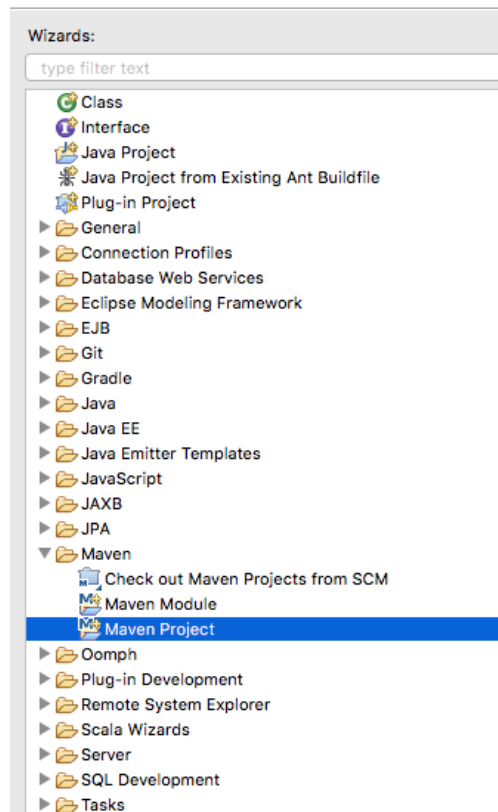


Eclipse: First Steps

- Close the Welcome View (and uncheck that you want to see this view next time)
- In the upper left, Click to make current the "Project Explorer" view
- Right-Click on whitespace in this view, and select New -> Other
- This action produces the modal dialog box on the right.
- Select Maven -> Maven Project -> Next

Select a wizard

Create a Maven Project



Eclipse: New Maven Project

New Maven project

Select project name and location

☐ Create a simple project (skip archetype selection)

☒ Use default Workspace location

Location:

☐ Add project(s) to working set

Working set:

► **Advanced**

Eclipse: Maven (Project type)

- Maven archetypes are essentially (project templates)
- Select " .. QuickStart", and Click, Next

New Maven project

Select an Archetype



Catalog: All Catalogs

Configure...

Filter:

Group Id	Artifact Id	Version
org.apache.maven.archetypes	maven-archetype-archetype	1.0
org.apache.maven.archetypes	maven-archetype-j2ee-simple	1.0
org.apache.maven.archetypes	maven-archetype-plugin	1.2
org.apache.maven.archetypes	maven-archetype-plugin-site	1.1
org.apache.maven.archetypes	maven-archetype-portlet	1.0.1
org.apache.maven.archetypes	maven-archetype-profiles	1.0-alpha-4
org.apache.maven.archetypes	maven-archetype-quickstart	1.1
org.apache.maven.archetypes	maven-archetype-site	1.1
org.apache.maven.archetypes	maven-archetype-site-simple	1.1
org.apache.maven.archetypes	maven-archetype-webapp	1.0

An archetype which contains a sample Maven project.

☒ Show the last version of Archetype only

☐ Include snapshot archetypes

Add Archetype...

► Advanced



< Back

Next >

Cancel

Finish

New Maven project

Specify Archetype parameters



Group Id:

Artifact Id:

Version:

Package:

Properties available from archetype:

Name	Value

► Advanced



< Back

Next >

Cancel

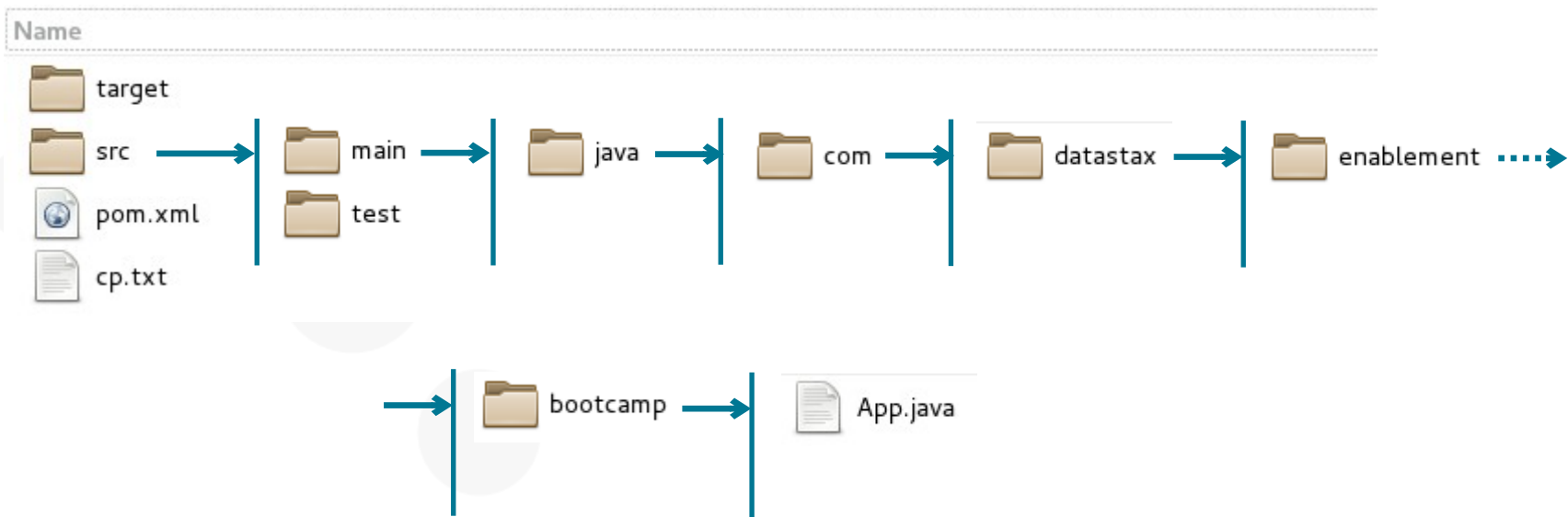
Finish

Group Id:

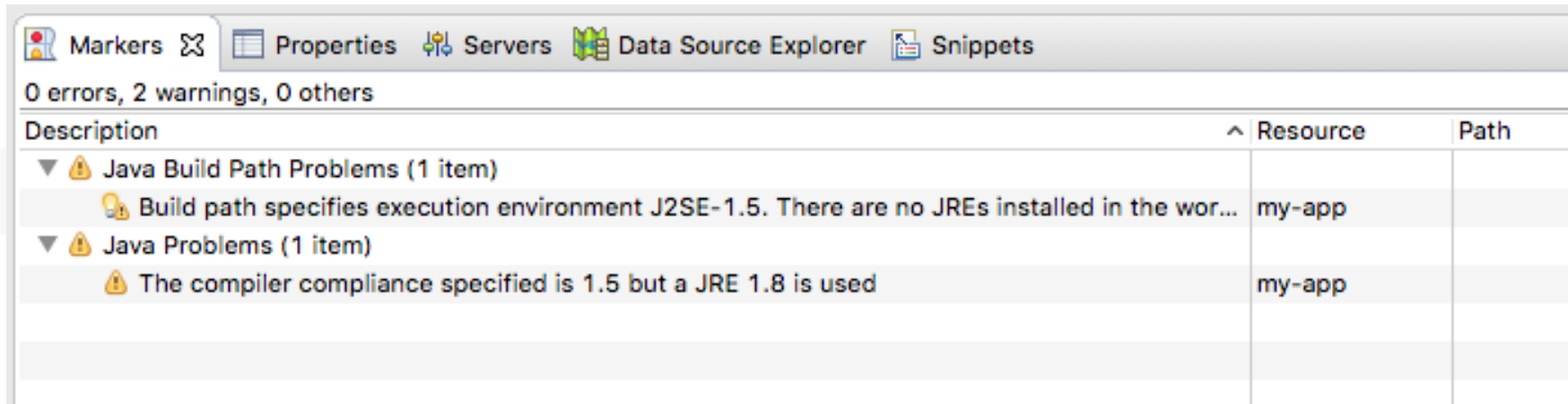
Artifact Id:

Eclipse: What just happened ?

Group id=`com.datastax.enablement.bootcamp`



Eclipse: Problem w/ Compiler Version

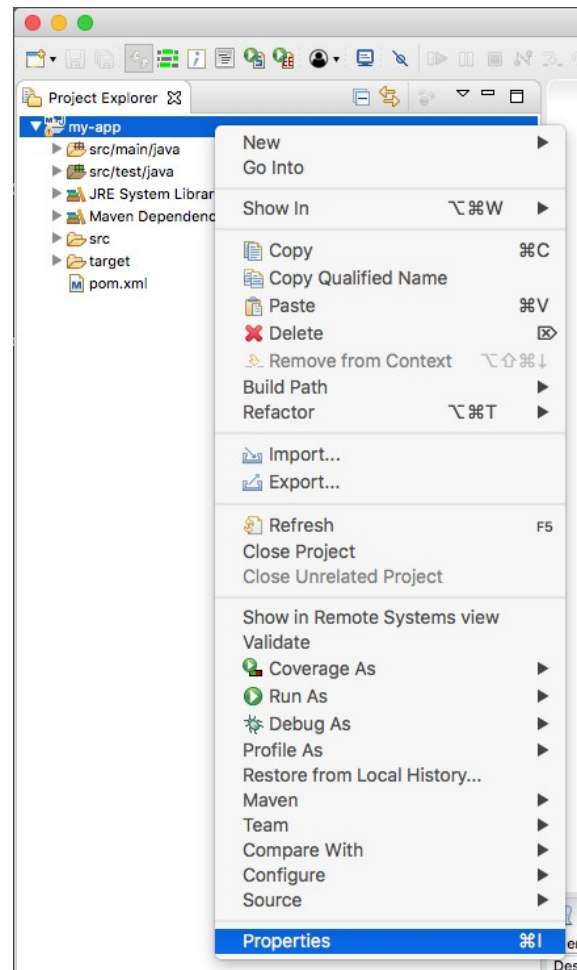


The screenshot shows the Eclipse IDE's 'Markers' view. At the top, it says '0 errors, 2 warnings, 0 others'. Below this, there are two expandable sections: 'Java Build Path Problems (1 item)' and 'Java Problems (1 item)'. The first section contains a warning icon and a message: 'Build path specifies execution environment J2SE-1.5. There are no JREs installed in the wor...'. The second section contains a warning icon and a message: 'The compiler compliance specified is 1.5 but a JRE 1.8 is used'. Both messages are associated with the resource 'my-app'.

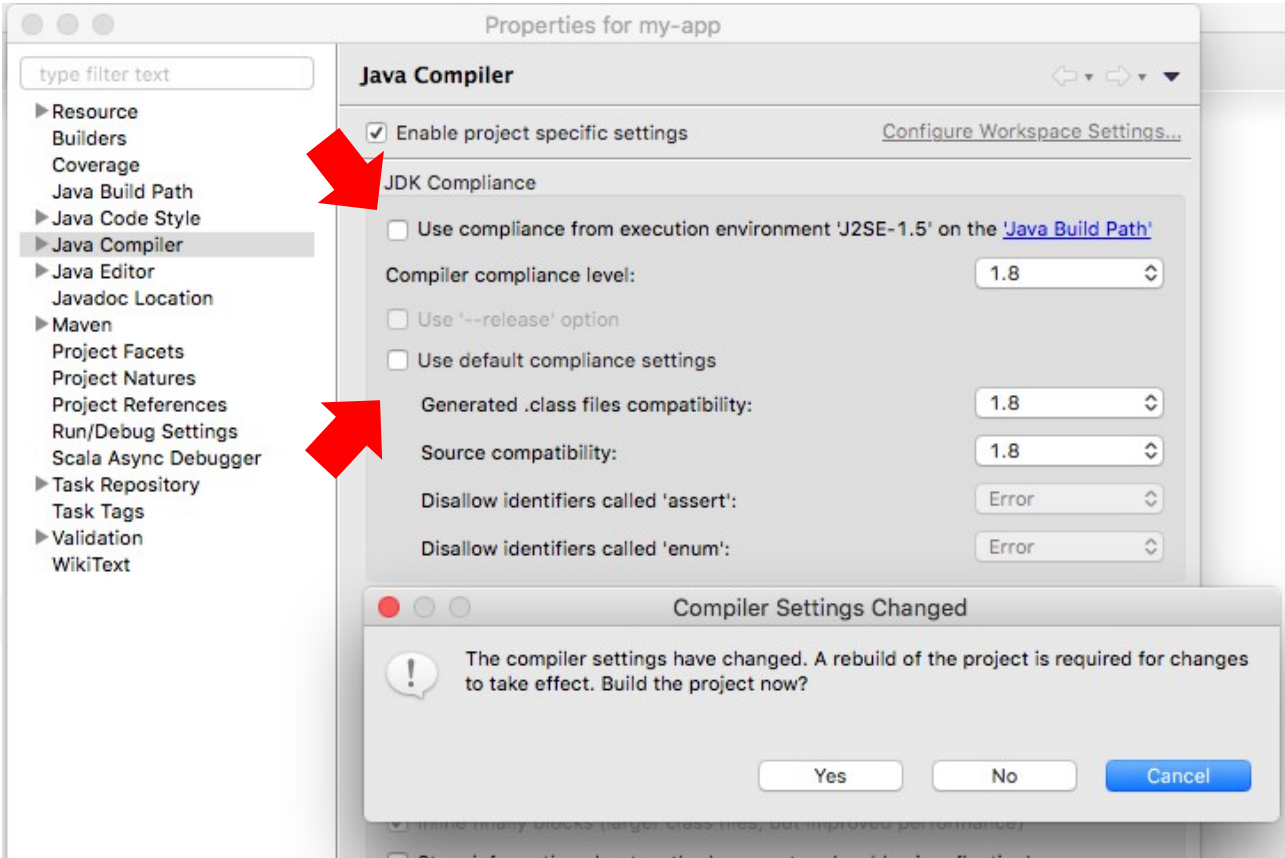
Description	Resource	Path
▼ ⚠ Java Build Path Problems (1 item)		
💡 Build path specifies execution environment J2SE-1.5. There are no JREs installed in the wor...	my-app	
▼ ⚠ Java Problems (1 item)		
⚠ The compiler compliance specified is 1.5 but a JRE 1.8 is used	my-app	

Eclipse: Changing Compiler Version-

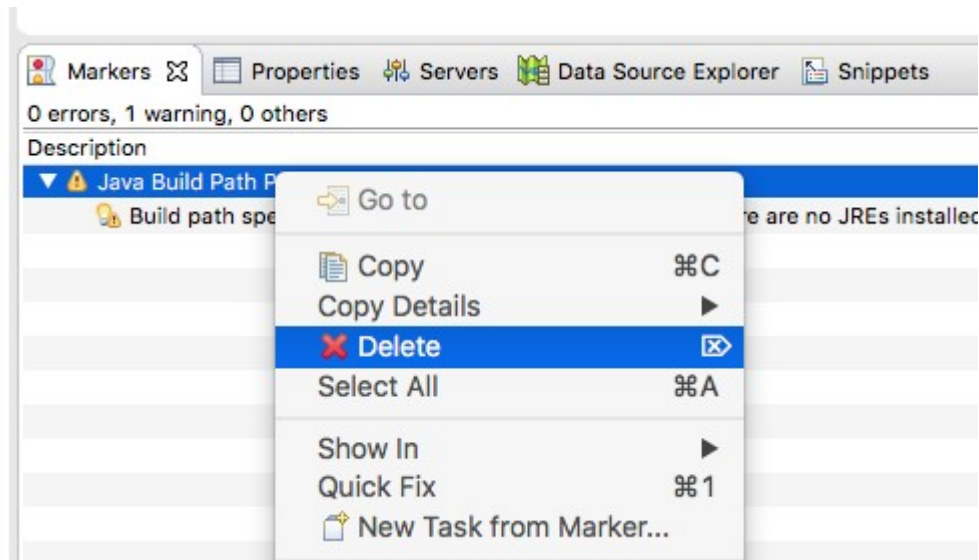
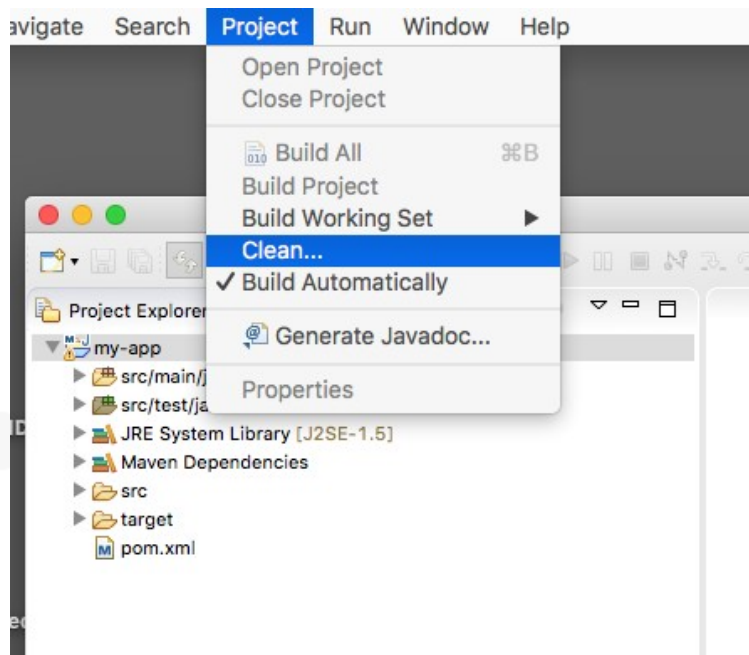
- In the Project Explorer view, Right-Click the project name (my-app)
- And select, Properties



Eclipse: Version 1.8 JDK



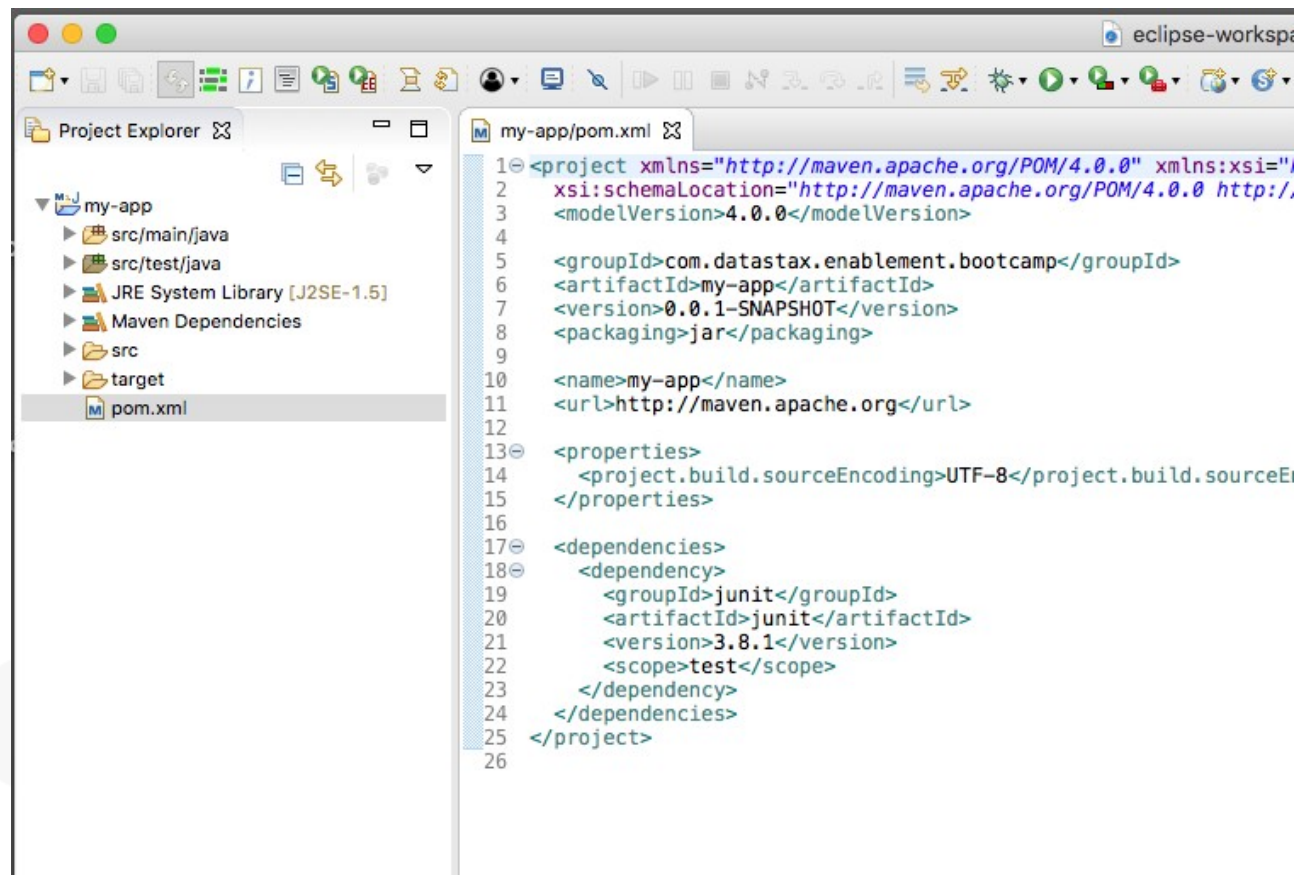
Eclipse: Clean, and (clean)





Ready to Program Java and a DSE Client

Project Explorer view, pom.xml



What is a pom.xml ?

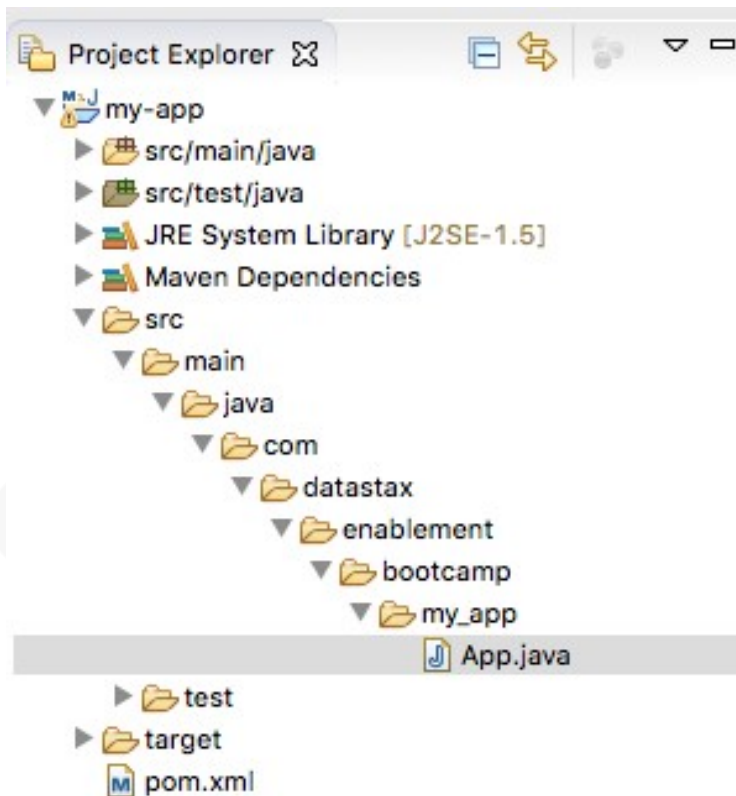
- xml
- "project object model"
- Each project has one
- Tells Maven how to compile, etc, your project
- Edit once and done ?

```
<project xmlns="http://maven.apache.or ...  
  xsi:schemaLocation="http://mav ...  
  <modelVersion>4.0.0</modelVersion>  
  
  <groupId>com.datastax.enablement.bootcamp<  
/groupId>  
  <artifactId>my-app</artifactId>  
  <packaging>jar</packaging>  
  <version>1.0-SNAPSHOT</version>  
  <name>my-app</name>  
  <url>http://maven.apache.org</url>  
  <dependencies>  
    <dependency>  
      <groupId>junit</groupId>  
      <artifactId>junit</artifactId>  
      <version>3.8.1</version>  
      <scope>test</scope>  
    </dependency>  
  </dependencies>  
</project>
```

A POM that supports DSE/Java-

- Full/complete POM pasted below on Notes page
- But in effect, only need this/additional "dependency" block
- Jar files pulled, by default, from an online Maven repository
- Also pulls whatever Jars these Jars need, etc
- This repository can be (should be) internal for real/production apps

```
<dependencies>
  <dependency>
    <groupId>com.datastax.dse</groupId>
    <artifactId>dse-java-driver-core</artifactId>
    <version>1.6.7</version>
  </dependency>
</dependencies>
```



Changing App.java

- In the Project Explorer view, navigate to App.java, and Double-Click to open this file in the Editor view

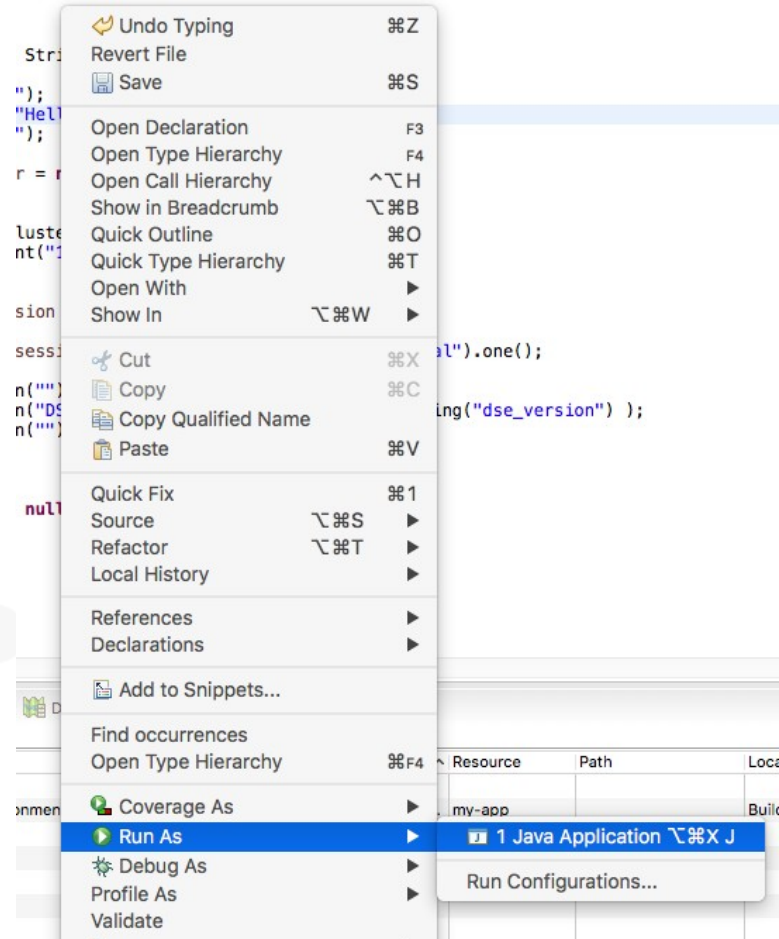
Changing App.java

- In the Editor view, change App.java to equal the source code pasted on the Notes page below.

```
my-app/pom.xml  *App.java  X
2
3 package com.datastax.enablement.bootcamp.my_app;
4
5 import com.datastax.driver.dse.DseCluster;
6 import com.datastax.driver.dse.DseSession;
7 //
8 import com.datastax.driver.core.Row;
9
10 /**
11  * Hello world!
12  *
13  */
14 public class App
15 {
16     public static void main( String[] args )
17     {
18         System.out.println("");
19         System.out.println("Hello World!");
20         System.out.println("");
21
22         DseCluster my_cluster = null;
23
24         try {
25             my_cluster = DseCluster.builder()
26                 .addContactPoint("127.0.0.1")
27                 .build();
28
29             DseSession my_session = my_cluster.connect();
30
31             Row my_row = my_session.execute("select * from system.local").one();
32
33             System.out.println("");
34             System.out.println("DSE release version: " + my_row.getString("dse_version") );
35             System.out.println("");
36
37         } finally {
38
39             if (my_cluster != null) my_cluster.close();
40
41         }
42
43     }
44
45 }
46
```


Eclipse: Compile and Run

- Anywhere in the Editor view (in App.java), Right-Click and select, Run As -> Java Application
- You will be prompted to save the changes you made to pom.xml, and App.java, so save



Eclipse: Success



The screenshot shows the Eclipse IDE's Console window. The title bar includes tabs for Markers, Properties, Servers, Data Source Explorer, Snippets, and Console. The console output shows the application has terminated, displaying 'Hello World!' in black text. Below this, there are three lines of red text indicating an SLF4J warning: 'SLF4J: Failed to load class "org.slf4j.impl.StaticLoggerBinder".', 'SLF4J: Defaulting to no-operation (NOP) logger implementation', and 'SLF4J: See http://www.slf4j.org/codes.html#StaticLoggerBinder for further details.'. At the bottom, it shows 'DSE release version: 6.0.0' in black text.

```
<terminated> App [Java Application] /Library/Java/JavaVirtualMachines/jdk1.8.0_171.jdk/Contents/Home/bin/java (Jul 2, 2016)

Hello World!

SLF4J: Failed to load class "org.slf4j.impl.StaticLoggerBinder".
SLF4J: Defaulting to no-operation (NOP) logger implementation
SLF4J: See http://www.slf4j.org/codes.html#StaticLoggerBinder for further details.

DSE release version: 6.0.0
```

Practice Lab:

Lessons Learned





End of Unit: